

aerial intelligence that runs on **your desk**.

three vision models stitched into one pipeline – tracking,
grounding, reasoning – over drone and camera footage, entirely on
apple silicon. no cloud round-trip. no per-seat pricing. no
footage leaving the building.

● SAM 3.1 · TRACKING

● FALCON PERCEPTION · GROUNDING

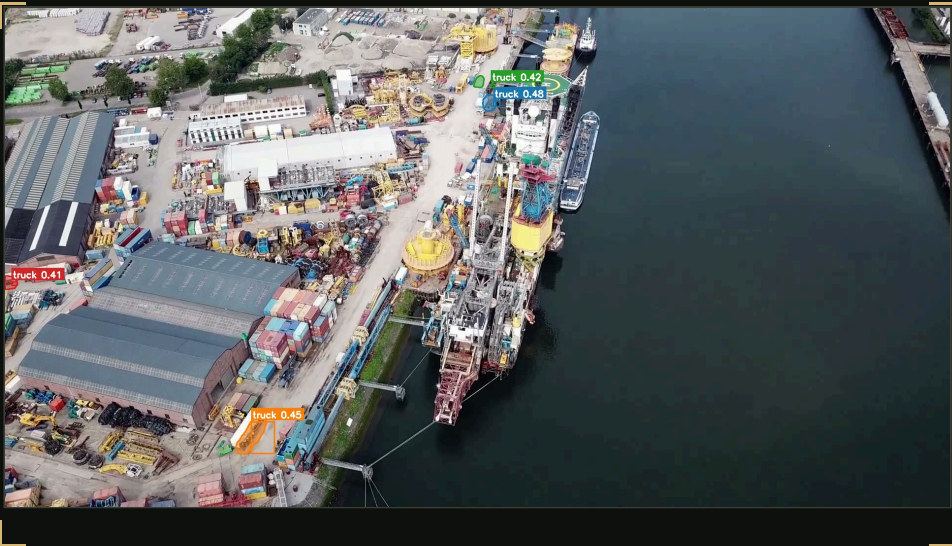
● GEMMA 4 · REASONING

one query in. three models later: annotated video, structured json, and a written field report.

type "vessels and trucks at the quay" – the prompt router splits your sentence into tracking targets for sam 3.1 and a semantic question for gemma. the output is the triad ops teams actually need.

1 • TRACKING	SAM 3.1	per-frame masks and persistent track ids across the whole clip, on-device via MLX.
2 • GROUNDING	Falcon Perception	expression-level detect, segment, and ocr on key frames – "person pushing a stroller", not just "person".
3 • REASONING	Gemma 4	counts, anomalies, and recommended actions in a field report – resolved locally via ollama, your own gpu server, or on-device.

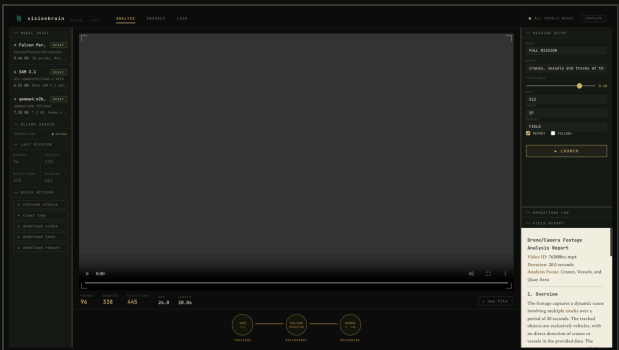
every sweep becomes **evidence**: the clip, the counts, and the write-up.



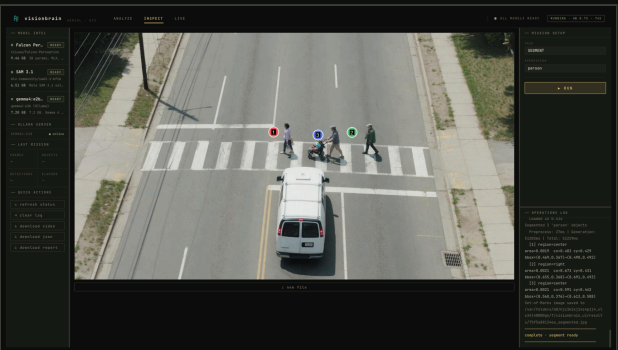
LEFT — MISSION VIEW: 480 FRAMES → 96 PROCESSED • 338 OBJECTS • 445 DETECTIONS • FIELD REPORT IN 66 S
RIGHT — ACTUAL ANNOTATED OUTPUT: SAM 3.1 TRACK CONTOURS + CONFIDENCE, RENDERED BY THE PIPELINE

ONE PLATFORM • MANY ENVIRONMENTS

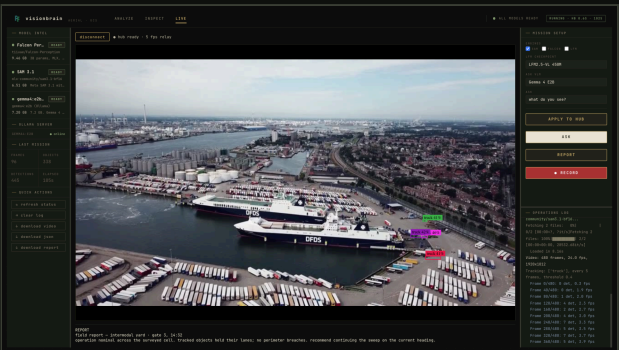
ports, yards, corridors – the same three questions:
what is here, what moved, what matters.



ANALYZE
FULL MISSION • STATS + REPORT



INSPECT
FALCON DETECT • SEGMENT • OCR



LIVE
FIELD HUB • 5 FPS RELAY

ask the scene a question. get an answer before the drone lands.

visionbrain

AERIAL · RCS · ANALYZE · INSPECT · LIVE

● ALL MODELS READY RUNNING · HB 0.85 · 405

MODEL INTEL

Falcon Per...
tiny/Falcon-Perception
9.46 GB 39 params, MLA, ...

SAM 3.1
mlx-community/sam3.1-bf16
6.51 GB Meta SAM 3.1 vit...

gemma4:c2b...
gemma4:c2b (D1Lena)
7.20 GB 7.2 GB, Gemma 4 ...

OLLAMA SERVER

GEMMA4:C2B online

LAST MISSION

FRAMES	OBJECTS
96	338
DETECTIONS	SLAPSED
645	105s

QUICK ACTIONS

refresh status

clear log

download video

download json

download report

disconnect ● hub ready · 5 fps relay

ANSWER

holding pattern over the quay. 4 tracked objects in frame, movement nominal. no perimeter events in the last 60 s.

MISSION SETUP

ENGINE

☒ SAM ☐ FALCON ☐ LPR

LPR CHECKPOINT

LFH2.5-VL 450M

ASK YLR

Gemma 4 E2B

ASK

what do you see?

APPLY TO HUB

ASK

REPORT

RECORD

OPERATIONS LOG

Memote Gemma:
http://100.72.41.118:8080
[1/3] SAM 3.1 - tracking ['truck']
in cc4065cd.mp4...
every 5 frames, backbone-
every 1, resolution=512,
threshold=0.4
Loading SAM 3.1 from mlx-
community/sam3.1-bf16...
Fetching 2 files: 0%
0/2 [00:00:07, 71k/s] Fetching 2
files: 100% [00:00:00, 28532.6k/s]
Loaded in 8.16s
Video: 480 frames, 24.0 fps,
1920x1012
Tracking: ['truck'], every 5
frames, threshold 0.4
Frame 0/480: 0 det, 0.3 fps
Frame 40/480: 0 det, 1.9 fps

OPERATOR
what do you see?

VISIONBRAIN · GEMMA 4 E2B

holding pattern over the quay. 4 tracked objects in frame, movement nominal. no perimeter events in the last 60 s.

VISIONBRAIN ♦

LIVE OPS · 05

measured, not projected. one 32 GB mac, all three models co-resident.

338 OBJECTS TRACKED · 20 S QUAY CLIP · ONE QUERY	~7 FPS END-TO-END TRACKING AT 512 PX	23.1 GB FALCON + SAM + GEMMA RESIDENT IN RAM	0 BYTES OF FOOTAGE TO ANY CLOUD
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zone counters (line + polygon), 8-way direction classification, and bytetrack identity come standard. fastscan answers "is it there?" in under a minute; the full pipeline keeps working behind it. chunked processing chews through long recordings that would starve a single-pass pipeline.

shipped, in progress, and **direction** – labeled exactly like that.

◆ SHIPPED – RUNNING TODAY

→ **full pipeline** – sam 3.1 track → falcon refine → gemma report, from one query

→ **ground control ui** – analyze · inspect · live panes, resizable operator layout

→ **live field hub** – relay, detection hud, ask-the-scene, reports, annotated recording

→ **fastscan** – quick answers in under a minute; full pipeline continues behind it

→ **structured video** – chunked processing for long recordings, adaptive sampling

→ **api + cli** – visionbrain detect | segment | ocr | track | analyze | fastscan | status

◇ IN PROGRESS · THEN DIRECTION

→ **fast-path hardening** – quick-answer-first, full pipeline behind it (spec'd, flags live in the ui)

→ **alert rules on zones** – dwell, loiter, wrong-way, perimeter breach → webhook on event

→ **hub federation** – one dashboard, n field relays, per-feed engine control

→ **thermal + low-light profiles** – night patrols without retuning queries

→ **scheduled sweeps** – waypoint missions with per-leg queries

→ **report templates per vertical** – port ops, terminals, corridors, perimeters

WHO THIS IS FOR

businesses that fly drones and can't ship the footage to someone else's cloud.

PORTS & TERMINALS berth oversight without shipping footage ashore. vessels, cranes, and truck traffic from one query.	LOGISTICS YARDS gate-to-gate truck counts, dwell time, and lane discipline – each a single question.
CORRIDORS & INFRASTRUCTURE every overpass, every sweep, written up before the drone lands.	SECURITY & PERIMETER live hub + zones + direction tracking: a fence that writes its own log.

```
pip install -e . → visionbrain status → open ground control at 127.0.0.1:7860 · bring a 20-second clip, leave with a report.
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